

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping production RAG, agentic LLM workflows, and evaluation-driven pipelines for healthcare and enterprise clients - ~35% retrieval precision gains, >30% hallucination reduction via grounded evals, ~25% agentic-workflow stability uplift, and ~30% post-deployment defect reduction. Hands-on with LangChain, LlamaIndex, FastAPI, Docker, scikit-learn, XGBoost, and Hugging Face Transformers. Comfortable owning end-to-end delivery from data through eval harnesses to deployed APIs, and translating model behavior to non-technical stakeholders in regulated environments.

CORE COMPETENCIES

RAG pipelines & grounded retrieval • Agentic LLM workflows • LLM evaluation & observability • LangChain / LlamaIndex • FastAPI & Docker • Structured outputs & schema validation • Technical advisory • HIPAA-conscious delivery

WORK EXPERIENCE

Progress Solutions Inc. - Jr. AI/ML Engineer Trainee

Jul 2025 - Present

USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- **Retrieval-Augmented Generation (RAG) pipelines** with evaluation-driven precision tuning: recursive semantic chunking + transformer-based embeddings + grounded eval harnesses raised contextual retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**, and lifted retrieval-grounded response alignment **above 90%**.
- **Agentic LLM workflows** for multi-step clinical and enterprise queries with structured reasoning, tool discipline, and grounding rules; improved agent response stability ~**25%** through schema-validated outputs and instrumented evaluation traces.
- **Predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall **15-20%** on high-risk categories while holding precision >**90%** via class weighting, stratified sampling, and threshold calibration.
- **LLM and ML inference services** packaged as FastAPI/Flask REST APIs, containerized with **Docker**, with structured logging, load simulation, and observability hooks; reduced post-deployment defects ~**30%**.
- **HIPAA-conscious data governance**: de-identification, lineage docs, evaluation audit trails, and stakeholder-facing system-limitation documentation that translated eval results to non-technical clinical leads.

Energy Solutions International (Emerson) - Database & DevOps Performance Engineer (Intern)

Jun 2022 - Apr 2023

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform (contracts, nominations, allocations, invoicing); reduced query execution time ~**20%** and data retrieval latency ~**25%**.
- **Jenkins CI/CD pipelines** for schema updates, version-controlled SQL scripts, and stored procedure deployments; reduced deployment errors >**30%** and improved release cycle ~**35-40%** via structured release validation and rollback checkpoints.
- Performance dashboards using **SQL Server DMVs and Grafana** to track long-running queries, deadlocks, and CPU/I/O contention; cut incident recurrence ~**25%** and improved root-cause resolution speed ~**30%**.
- Tuned database maintenance (index rebuilds, statistics updates, partition-aware indexing) and reconciliation queries; improved reconciliation runtime ~**15%** and reduced deadlocks ~**15%**.
- Supported **RBAC implementation and audit logging** for financial modules in a compliance-sensitive oil & gas environment.

Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; transcript preprocessing, hierarchical summarization, and timestamp-aligned clip extraction reduced manual review time **60-70%** (hours → <5 minutes per video).
- Achieved **~85% alignment** between AI-selected highlights and human-curated key moments.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs **~30%** and raised contextual answer accuracy **>85%**.
- Deployed as a **Flask-based API** with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** preventing cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app that interprets dreams, generates poetic reinterpretations, and synthesizes dreamscapes via Perplexity Sonar (interpretation) + Replicate (16:9 visuals). **Intermediate structured prompt transformation layers** lifted contextual alignment **~30%** and first-pass image success rate **~25-30%** over naive direct-prompt orchestration.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation (Shipped, Chrome Web Store)

Manifest V3 extension built solo - content scripts, service workers, **captureVisibleTab** for panel capture. **Four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) behind one abstraction. Multi-section pipeline for tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selectors for 29 manga/webtoon sites, privacy policy, and narrowed host permissions.

Agentic Pixel Character Synthesis & Animation Engine (ongoing)

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning, latent-space consistency for animation smoothness, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

WORK AUTHORIZATION

F-1 OPT - Authorized to work in the United States; future sponsorship required.

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, responsible AI

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers & Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems